Code No: 135BY **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
COMPUTER GRAPHICS
(Common to CE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	How does video display work?	[2]
b)	Explain the architecture of a raster-graphics system with a display processor.	[3]
c)	How to perform general pivot point rotation and general fixed point scaling?	[2]
d)	Discuss the composite transforms.	[3]
e)	Briefly describe the sweep representation of 3D objects.	[2]
f)	List the types of 3D object representation.	[3]
g)	How wireframe displays might be generated with the various visible-surface detection methods?	[2]
h)	What are the advantages of Backface detection method?	[3]
i)	What is animation in computer graphics PDF?	[2]
j)	Give a brief note on raster animation.	[3]

PART – B

(50 Marks)

2.a) Describe the Scan line polygon fill algorithm.
b) Explain the random scan systems. [5+5]

OR

3.a) List and explain the application areas of Computer Graphics.
b) What are Graphic primitives? Mention some typical graphic primitives that a package may provide. [5+5]

4. Explain the Cohen-Sutherland line clipping algorithm with an example. [10]

OR

5. Derive the 2D transformation matrix for rotation about an arbitrary point. [10]

6. Explain the 3D reflection and shear transformations. [10]

OR

7. Derive the transformation matrix for Hermite Curve. [10]

8. Write a program to display the visible surfaces of a convex polyhedron using the BSP-tree method. [10]

OR

9. Illustrate the two image space methods that are used to deal with visible surface detection. [10]

10.a) Explain the computer animation languages.
b) Give a brief note on keyframe systems. [5+5]

OR

11. Explain the general computer animation functions in detail. [10]

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